Course: Actuarial Science
Course Code: MIFA4152
Practical Term Paper Project Guidelines

Department of Modeling and Industrial Applications

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1 Project Objectives

The primary objectives of this practical term paper project are to:

1. **Synthesize Theoretical Knowledge:** Demonstrate a deep understanding of core life actuarial science, from primary survival models through to advanced pension mathematics.
2. **Develop Computational Skills:** Design and execute actuary-specific simulations (e.g., simulating mortality under different parametric laws, modeling fire claims) using modern statistical software (Python, R, or STATA).
3. **Apply Risk Models in Practice:** Evaluate reinsurance strategies for property/casualty risks (specifically fire claims) alongside traditional life insurance valuations.
4. **Professional Communication:** Draft a mathematically exact, properly formatted, and academically rigorous document using LaTeX/Overleaf.

2 Core Modules to be Addressed

Your practical project must integrate components acting upon the following six modules:

Module 1: Introduction to Life Actuarial Science & Survival Models
Establishing the foundation of mortality and survival distributions. Incorporate the parametric modeling of human mortality.

Module 2: Life Insurance and Annuity Functions
Modeling Expected Present Values (EPV) for various discrete and continuous insurance/annuity products.

Module 3: Premium Calculation & Risk Models
Developing net and gross premium formulations. *Specific Requirement:* You must incorporate and simulate the **Makeham Law (1860)** and the **Weibull Law**. Furthermore, this module must introduce Non-Life Risk Models by evaluating **Fire Claims for Reinsurance** (e.g., modeling claim severity/frequency and excess-of-loss treaties).

Module 4: Policy Values and Reserves

Projecting future liabilities. Calculating prospective, retrospective, and recursive reserves over the timeline of a policy.

Module 5: Multiple Life and Multiple Decrement Models

Modeling joint-life and last-survivor status. Incorporating models where a life can leave the observed group due to competing risks (e.g., death, withdrawal, disability).

Module 6: Pension Actuarial Science

Valuating retirement benefits, including defined benefit (DB) obligations, salary-scale functions, and standard actuarial cost methods (e.g., Projected Unit Credit).

3 Tasks Involved & Computational Requirements

Students must complete the following mandatory tasks within the paper:

1. **Simulation and Coding:**
    - Use **Python**, **R**, or **STATA** to simulate a cohort of at least 10,000 independent lives subject to Makeham and Weibull mortality laws.
    - Simulate a heavy-tailed distribution (e.g., Pareto or Lognormal) representing a portfolio of Fire Claims to evaluate a reinsurance treaty.
    - Code snippets and visual outputs (graphs of survival functions, reserve trajectories, and claim severity distributions) must be embedded in the text.
2. **Mathematical Derivations:** Provide step-by-step rigorous LaTeX derivations for at least one core theorem or reserve formula used in your analysis.
3. **LaTeX Formatting:** The final submission *must* be compiled via Overleaf, utilizing proper mathematical environments (`align`, `equation`), bibliographic citations (BibTeX), and formatted tables (`tabularx`).

4 Proposed Structure of the Term Paper

To ensure grading consistency and logical flow, the Term Paper should strictly adhere to the following outline:

1. Introduction

- Background of Actuarial Modeling.
- Clear statement of the paper's objectives.
- Overview of the computational methodology utilized.

2. Survival Models and Demographic Baselines (Module 1 & 2)

- Definition of the basic survival function, force of mortality, and life tables.
- Introduction of the life insurance and annuity formulations applied to the baseline cohort.

3. Parametric Risk Models and Pricing (Module 3)

- *Life Risks:* Fitting and simulating the Makeham Law and Weibull Law to the modeled population. Premium calculations under these laws.
- *Non-Life Risks:* Analysis of Fire Claims for a simulated P&C portfolio. Modeling severity/frequency and assessing reinsurance impacts.

4. Reserving and Liability Management (Module 4)

- Prospective calculation of policy values.
- Visualizing the trajectory of reserves over the policy duration via graphical simulations (Python/R/STATA output).

5. Advanced States: Multiple Decrements and Pensions (Module 5 & 6)

- Extending the single-life model to incorporate competing risks (e.g., death vs. disability).
- Constructing a basic pension valuation framework (Defined Benefit plan) using the projected decrement tables.

6. Simulation Results and Discussion

- Comprehensive presentation of all empirical findings from the Python/R/STATA models.
- Interpretation of the simulated fire claim reinsurance thresholds versus the life insurance mortality outcomes.

7. Conclusion and Practical Implications

- Summary of findings.
- Discussion on the integration between Life and Non-Life risk mathematics in modern actuarial portfolios.

8. References / Bibliography

- Standardized citations generated via BibTeX.

9. Appendix: Computational Scripts

- Full, commented source code (Python, R, or STATA `.do` files) used for the simulations, formatted using the `listings` or `minted` package.